

CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

# Chapter 2: Polynomials

Every formula and result of the chapter on one sheet

|    |                                 |                                                                           |
|----|---------------------------------|---------------------------------------------------------------------------|
| 1  | Linear polynomial               | $ax + b, a \neq 0; \text{zero} = -\frac{b}{a}$                            |
| 2  | Quadratic polynomial            | $ax^2 + bx + c, a \neq 0; \text{at most 2 zeroes}$                        |
| 3  | Cubic polynomial                | $ax^3 + bx^2 + cx + d, a \neq 0; \text{at most 3 zeroes}$                 |
| 4  | Sum of zeroes (quadratic)       | $\alpha + \beta = -\frac{b}{a}$                                           |
| 5  | Product of zeroes (quadratic)   | $\alpha\beta = \frac{c}{a}$                                               |
| 6  | Quadratic from its zeroes       | $k[x^2 - (\alpha + \beta)x + \alpha\beta], k \neq 0$                      |
| 7  | Sum of squares of zeroes        | $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$                  |
| 8  | Sum of reciprocals              | $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$ |
| 9  | Square of the difference        | $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$                  |
| 10 | Sum of cubes of zeroes          | $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$  |
| 11 | Cubic: sum of zeroes            | $\alpha + \beta + \gamma = -\frac{b}{a}$                                  |
| 12 | Cubic: sum of products in pairs | $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$                  |
| 13 | Cubic: product of zeroes        | $\alpha\beta\gamma = -\frac{d}{a}$                                        |
| 14 | Graph of a quadratic            | Parabola; opens upwards if $a > 0$ , downwards if $a < 0$                 |